

DANIEL TICAU

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Education

University of California San Diego

Sep. 2025 – May 2028 (expected)

Bachelor of Science in Computer Engineering — GPA: 3.9

La Jolla, California

- Relevant Coursework: Linear Algebra, Differential Equations, Multivariable Calculus, Discrete Mathematics, Object-Oriented Programming & Data Structures, Programming in C

Deep Learning Specialization – Coursera (Andrew Ng)

Nov. 2023 – Mar. 2024

- 5-course certification covering neural networks, CNNs, RNNs (GRUs/LSTMs), and transformers. Implemented models for speech transcription and obstacle detection using NumPy, Keras, and TensorFlow.

Experience

SDx at UC San Diego

Dec. 2025 – Present

Principal Member

- Organized 3 hackathons partnering with Anthropic, Helix DB, and Vercel, bringing together builders across the greater San Diego area to ship projects in single-day sprints
- Hosted 3 fireside chats with YC founders and a YC office hours session with 3 partners, connecting student teams with direct investor feedback on early-stage projects
- Ran 5 demo days where members publicly presented and received feedback on their projects, fostering a culture of building in the open

FIRST Tech Challenge – Deja Vu Robotics #13216

Sep. 2021 – Jan. 2025

Hardware Member / Driver

Santa Clara, CA

- Designed and fabricated aluminum mecanum wheel chassis with custom intake/outtake mechanisms using CAD software (Fusion 360, Onshape) and 3D printing (Ender 3) across 3 competition seasons
- Programmed autonomous navigation routines and tuned PID-based gamepad teleoperation controls in Java on Android Studio as primary chassis driver
- Awards: 2× Inspire, 1× Think, 1× Innovate, 1× Design

Projects

Splat – Hand-Gesture Physics Playground

Jan. 2025

- Built a real-time computer vision application that lets users draw shapes and interact with a 2D rigid-body physics environment through natural hand gestures via webcam, demonstrated live to 60+ attendees at SDx Hack Day
- Integrated MediaPipe Hands for multi-hand gesture recognition and Matter.js for physics simulation, implementing a finger-tracking state machine to support draw, grab, and throw interactions
- Won Most Creative Award by popular vote at Vercel Hack Day hosted by SDx at UC San Diego

Verifis – Low-Latency AI Fact-Checking Chrome Extension

Sep. 2025 – Dec. 2025

- Developed a full-stack Chrome Extension (Manifest V3) that lets users highlight any claim on a webpage and instantly surfaces credible sources supporting or refuting it
- Built a React/TypeScript frontend bundled with Vite and a RESTful Hono/Node.js backend API that integrates Google Gemini via AI Studio for NLP-based claim extraction, decomposition, and source retrieval
- Designed an asynchronous pipeline that parallelizes claim parsing and web source lookup to minimize end-to-end response latency

Technical Skills

Languages & Tools: Python, C, C++, Java, TypeScript, JavaScript, HTML/CSS, Git, GitHub, Linux, Docker, Jupyter Notebook, Google Colab, VS Code, Cursor, Claude Code

Frameworks & Concepts: PyTorch, TensorFlow, Keras, NumPy, pandas, scikit-learn, OpenCV, Neural Networks, Computer Vision, NLP, Data Structures & Algorithms, OOP, React, Node.js, Hono, Vite, MediaPipe, Matter.js, aiohttp, REST APIs

Miscellaneous

Silicon Valley Education Foundation (SVEF)

June 2025 – July 2025

Elevate Math Tutor

Los Arboles Literacy & Technology Academy

- Taught advanced math to 90 incoming 5th graders across 3 classrooms over a 5-week summer program to prepare them for the coming school year
- Led hands-on engineering projects (puff rockets, cardboard rovers) and delivered college readiness lessons to provide early STEM exposure